

Course: BCA

Subject: Database Management System

Subject Code: BCA-204

Time: 1.5 Hours

Max. Marks: 20

Instructions:- i) Attempt **THREE** questions in all.

ii) **Part A is compulsory**. Attempt any **FIVE** parts from it.

iii) Attempt **TWO** questions from Part B by selecting one question from each section.

PART A

Q1.

- i). Describe the Data Independence and its types.
- ii). Write five advantages of using a DBMS.
- iii). Difference between Primary Key and Foreign Key.
- iv). What are the different types of relationship in the Relational Model.
- v). What is a Candidate Key?
- vi). What is Normalization in database design?
- vii). Write a short note on Functional Dependency

(5 x 2 = 10)

PART B

Section I

Q2. Explain the DBMS with application.

Q3. Describe the ER Model and its Components. Discuss the importance of ER Model in database design and provide example of ER Design.

(5)

Section II

Q4. What is a Query Optimization and its importance.

(5)

Q5. What is a Functional Dependency and its types?

(5)

Entity, Relation, multivalued, Derived, trivial & non-trivial.